

Dear host,

Thank you for organising the event. It's an excellent event. We spent about two weeks working on this project. Due to time constraints, we haven't got time to write the full report. Instead, we hereby provide an overview of the method used. Please let us know if you need more details.

DoseRAD2026 Method Description

Dr. Yu Sun, Dr. Price Jackson

Peter MacCallum Cancer Centre, Melbourne, Australia

Overall strategy

- We assume the dose prediction behaviour is more uniform in the BEV hence we rotated all control points to the 0 degree gantry angle.
- We calculated the beam path (binary mask), by
 - First calculating the MLC shape on the isocentre slice.
 - Then the source location, which was treated as the focal point
 - The shape of MLC on each slice can then be interpolated / extrapolated based on the source location and the slice locations.
 - We first implemented this in SimpleITK then using torch grid sample to speed it up.
- Based on the BEV, we can calculate the water equivalent depth (WED), relative electron density (RED) and a few other features related to the beam-material interaction.
- Ideally we wanted to explore a model that captures behaviour of beams passing each slice, e.g. a 2.5D model along the beam path. Since speed is one major criterion and an MLP was chosen.
- The model works like an onion:
 - First predicting the core of the beam.
 - Then the peripherals, layer by layer in 3D.
 - This is because the correlation at the core with the RED, WED is very high.
 - A correct prediction of the core can help the peripherals.
 - There is one core and three layers in this case.
 - The prediction of the core acts as an input for the close-by peripheral.

We used 50 patients as training, 15 as validation and 10 for testing.

The final model resulted in an L2 loss of 5×10^{-6} on the dose file.